

# Industry AI Flow

AI-powered construction intelligence — search building codes, predict cost overruns, and analyze project data with full privacy.

Angel Bustamante | Jason Niu | Jack Si

Instructor: Reeta Suman · SAIT

## 🚫 The Pain Point

Construction teams spend **hours manually searching** through thousands of pages of building codes, safety manuals, and specs. Answers are slow, scattered, and error-prone.

## ✅ Our Solution

VS

A private **AI-powered platform** that understands construction documents, predicts cost overruns with ML, and generates data insights — all running on local hardware for complete privacy.

## 1 Three Core Capabilities



PRIMARY FEATURE

### RAG Knowledge QA

Upload building codes and ask questions in natural language. Hybrid search (BM25 + pgvector + RRF) with cross-encoder reranking and cited answers.



ML + EXPLAINABILITY

### Cost Estimation

CatBoost + Ridge dual model on 10K remediated projects. SHAP TreeExplainer shows top-5 cost drivers. What-if scenarios and similar project lookup.



CODE GEN + SSE

### Dynamic Data Analysis

Upload any dataset — merged AI prompt generates analysis + visualization in one call. 6-node SSE pipeline with PII detection and Docker/E2B sandbox.

## 2 Product Architecture

Three AI capabilities powered by one intelligent router

11-Node Intent Router

Input

→

Classify

→

Confidence

→

Route



PRIMARY FEATURE

### RAG Knowledge QA

- 1 Hybrid Search (BM25 + pgvector)
- 2 Cross-Encoder Reranking
- 3 LLM Cited Generation

#### Cited Answers

Source references + groundedness check



ML + EXPLAINABILITY

### Cost Estimation

- 1 CatBoost + Ridge Dual Model
- 2 SHAP Top-5 Cost Drivers
- 3 What-If + Similar Projects

#### Cost Intelligence

Overrun prediction + SHAP waterfall



SSE PIPELINE

### Data Analysis

- 1 PII Detect + Metadata Extract
- 2 Cloud LLM Code Generation
- 3 Docker Sandbox Execution

#### Charts + Insights

6-node SSE real-time streaming

Privacy by Design

PII Detection + Metadata-Only Cloud

Local LLM (Ollama Metal GPU)

Docker + E2B Sandbox

10-NODE EXECUTION PIPELINE — fixed-order workflow processing every query



Intent

245ms

→



Safety

12ms

→



Cost

85ms

→



Retrieval

1.65s

→



Rerank

340ms

→



Prompt

15ms

→



Route

8ms

→



Execute

~15s

→



Response

120ms

→



Verify

45ms

28

Construction Documents Indexed

10K

Training Projects  
BCPI Remediated Dataset

11+10

Node Pipeline  
Intent + Workflow Graphs

0.993

R<sup>2</sup> Cost Accuracy  
Ridge Regression Model

## 3 System Architecture

Four-layer stack from frontend to data



Frontend

Next.js 16 · App Router

Workflow Chat

Cost Estimation

Data Analysis

Documents

Intent Debugger

Dashboard

▼ ▼ ▼



API + Security

FastAPI · JWT

JWT Auth

Tenant Isolation

Rate Limiter

Audit Logger

Input Sanitizer

▼ ▼ ▼



AI Engine

LangGraph · StateGraph

Intent Classifier · 11-Node

Pipeline · 10-Node

RAG · Hybrid Search

CatBoost + SHAP

Code Executor

▼ ▼ ▼



Data + LLM

PostgreSQL · pgvector

pgvector · 768-dim

Ollama · Qwen3.5 · GPU

Cloud LLM · Zhipu / Groq

bge-reranker

User Query

→

Next.js

→

FastAPI

→

Auth

→

Intent (11-Node)

→

Pipeline (10-Node)

→

Response + Citations

### NBC 2020

National Building Code of Canada — fire safety, structural, plumbing

### Ontario Reg. 213/91

Construction safety — fall protection, scaffolding, excavation

### BC Building Code

2024 edition — seismic, energy efficiency, accessibility

### Canada OHS

Occupational health & safety — employer obligations, PPE

### Labour Code II

Canada Labour Code Part II — employee rights, work refusal

### Quebec Safety

Quebec construction safety code — provincial requirements

FastAPI

LangGraph

PostgreSQL + pgvector

Ollama (Qwen 3.5)

Next.js 16

Docker + E2B

CatBoost + SHAP



Try the Demo  
lai.dissidia.me



Angel Bustamante  
LinkedIn



Jason Niu  
LinkedIn



Jack Si  
LinkedIn